Synopsis

Project Title: Employee Salary Prediction using Machine Learning

Problem Id: 3

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- **Problem Description:**

This project focuses on developing a regression-based machine learning system to predict employee salary using a synthetically generated dataset. The dataset simulates a real organizational environment and includes multiple employee-related features such as age, experience, education level, job role, performance rating, and skill score. Salary prediction is important for organizations to ensure fair compensation, improve employee satisfaction, and support decision-making in human resource management. The system analyzes relationships between different employee attributes and salary using statistical and machine learning techniques. By applying regression models, the project aims to accurately estimate salaries based on input features. This approach helps in understanding key factors influencing salary and demonstrates the practical application of machine learning in real-world business problems.

- **Steps of Implementation:**

- Data Generation: Create synthetic dataset with 20 features.
- Data Loading: Load dataset using pandas.
- EDA: Analyze data using graphs and statistics.
- Preprocessing: Handle missing values, encoding, scaling.
- Feature Selection: Apply forward selection.
- Model Training: Linear, Ridge, Lasso Regression.
- Evaluation: Use R^2 and RMSE.
- Result Analysis: Compare models.

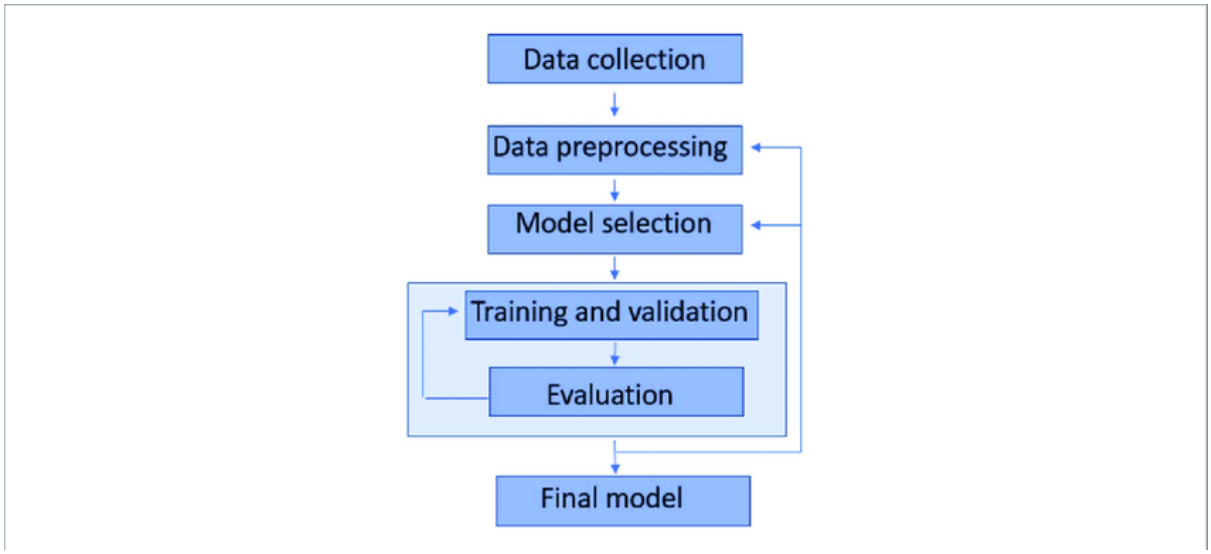

Fig. 1 Architecture

- **Expected Modules/ Libraries to be used:**

- **Pandas** -> Used for loading, handling, and manipulating the dataset.
- **NumPy** -> Used for numerical operations and working with arrays.
- **Matplotlib ->** Used to create basic graphs and visualizations.
- **Seaborn ->** Used for advanced and attractive statistical plots.
- **Scikit-learn** -> Used for preprocessing, feature selection, model training, and evaluation.

- **Expected Evaluation metrics and Plots to be used:**

1. **Evaluation metrics:**

- **R^2 Score:** Measures how well the model explains the variance in salary (closer to 1 is better).
- **RMSE (Root Mean Square Error):** Measures the average prediction error (lower is better).

2. **Plots:**

- **Histogram:** Used to understand the distribution of salary and other features.
- **Correlation Heatmap:** Shows relationships between features and salary.
- **Scatter Plot (Actual vs Predicted):** Used to compare real and predicted salary values.

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- **Required topics from the subject:**

- Data Structures
- File Handling
- Functions

- **Platform Required:**

Jupyter Notebook / Python

- **Books and Link Sources:**

1. **Hands-On Machine Learning with Scikit-Learn** – Aurélien Géron
2. **Introduction to Machine Learning with Python** – Andreas Müller
3. **Scikit-learn Official Documentation** – https://scikit-learn.org
4. **Kaggle** – https://www.kaggle.com